JMORF/MORS — Morpho-Syntax

Sign-Based Construction Grammar

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Lecture 12
Location: KB-1.11

Overview

Chapter 16 framework (same analyses, different underlying system)

- Construction grammar
  - The construction grammar family
  - Differences between SBCG and HPSG
- Multiple inheritance: lexemes and clauses
- Everything is a sign: words, rules, principles and the root condition

Construction Grammar

Construction Grammar (CxG)

- A family of grammars based on the idea that
  - Knowledge of a language comes from form and function pairings
  - **function** includes meaning, content, or intent (both semantics and pragmatics)
  - **form** includes phonology, syntax, orthography

- CxG grew out of generative semantics and cognitive linguistics, by researchers such as Charles Fillmore, Paul Kay and George Lakoff

- Instead of language as a grammar+lexicon, think of it as a structured network of families of constructions

Construction Grammars

- **Sign-Based Construction Grammar** (Berkeley Construction Grammar)
  unification-based framework (with computational implementation)

- **Goldbergian/Lakoffian Construction Grammar**
  psychologically plausible

- **Radical Construction Grammar**
  syntactic categories, roles, and relations are not universal:
  they are not only language-specific, but also construction-specific

- **Embodied Construction Grammar**
  relates constructions to embodiment and sensorimotor experience

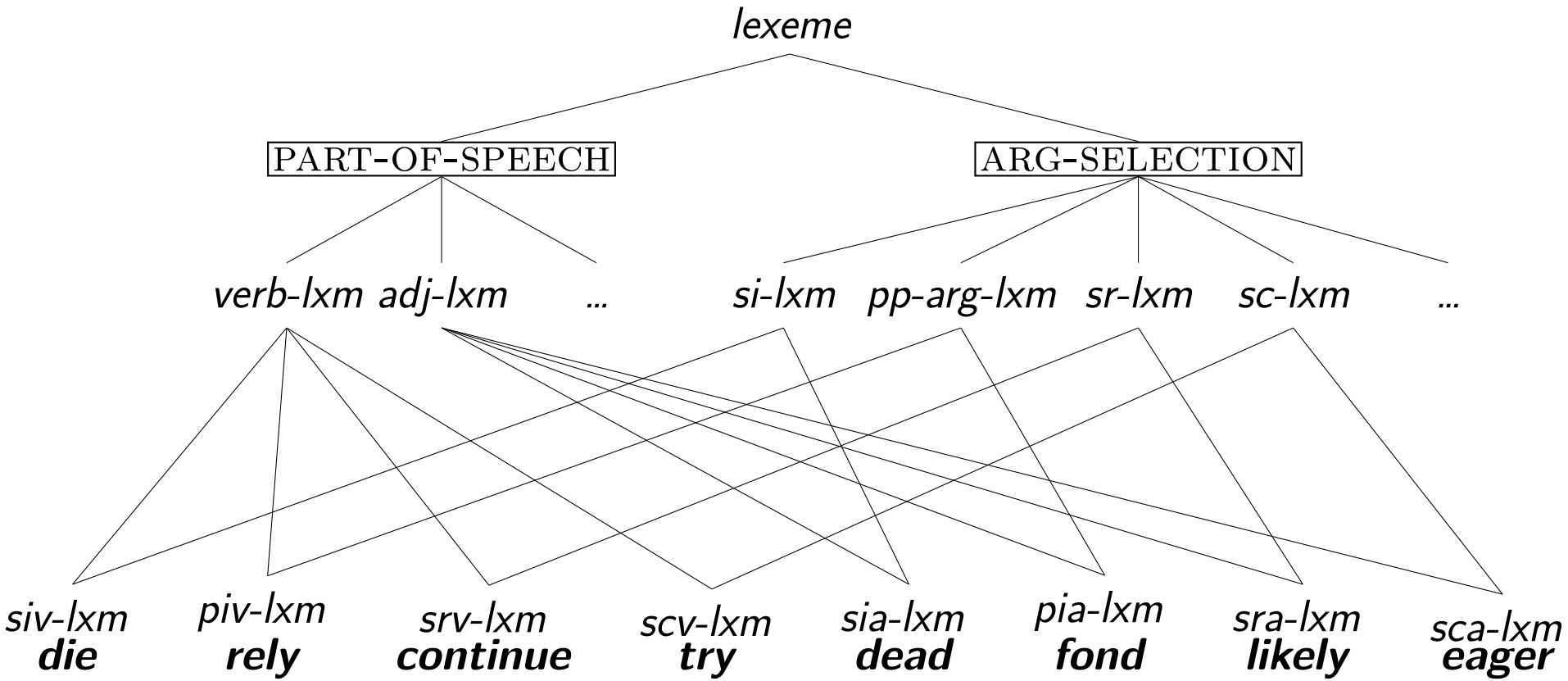
- **Fluid Construction Grammar**
  learns grammars from the environment (with computational implementation)

Overview of Differences (SBCG vs. HPSG)

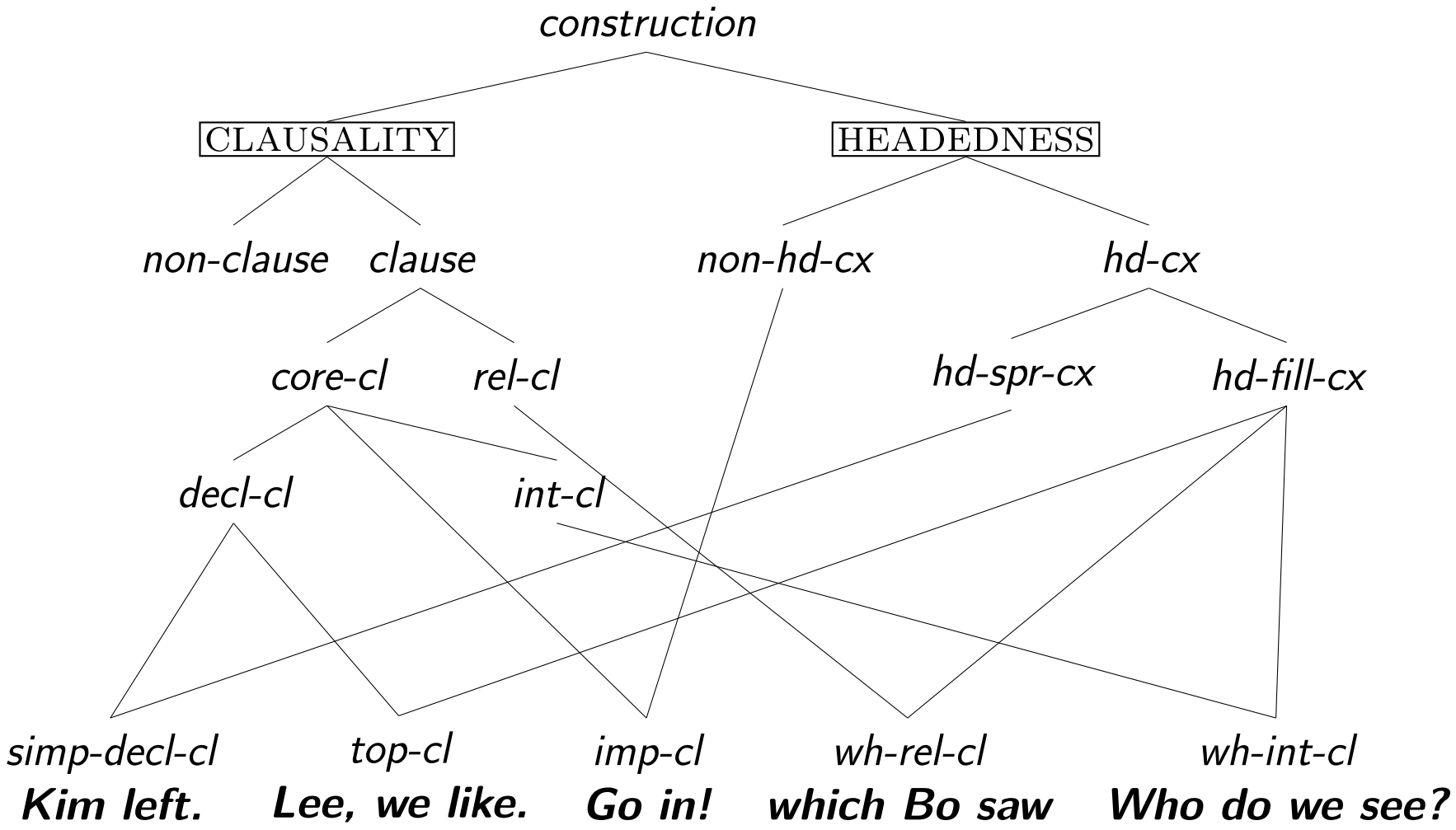
- Multiple inheritance
- Signs
- Grammar rules form a hierarchy — many more rules
- Every tree node has its own phonology
- Many principles become constraints on grammar rules
- The definition of well-formedness is simplified

Multiple Inheritance

More Inheritance (Lexemes)



More Inheritance (Clauses)



Everything Is a Sign

Words Are Signs

$$\left\langle \text{dog}, \begin{bmatrix} \textit{word} \\ \text{HEAD} \;\; \textit{noun} \\ \text{VAL} \;\; \begin{bmatrix} \text{SPR} \;\; \langle \text{D} \rangle \\ \text{COMPS} \;\; \langle\rangle \end{bmatrix} \end{bmatrix} \right\rangle$$

Becomes:

$$\begin{bmatrix} \textit{word} \\ \text{ORTH} \;\; \langle \text{“dog”} \rangle \\ \text{SYN} \;\; \begin{bmatrix} \text{HEAD} \;\; \textit{noun} \\ \text{VAL} \;\; \begin{bmatrix} \text{SPR} \;\; \langle \text{D} \rangle \\ \text{COMPS} \;\; \langle\rangle \end{bmatrix} \end{bmatrix} \end{bmatrix}$$

Rules Are Signs

$$\begin{bmatrix} \textit{phrase} \\ \text{VAL} \begin{bmatrix} \text{COMPS} & \langle\rangle \\ \text{SPR} & \langle\rangle \end{bmatrix} \end{bmatrix} \rightarrow \boxed{2} \quad \mathbf{H} \begin{bmatrix} \text{VAL} \begin{bmatrix} \text{COMPS} & \langle\rangle \\ \text{SPR} & \langle \boxed{2} \rangle \end{bmatrix} \end{bmatrix}$$

Becomes:

$$\begin{bmatrix} \textit{head-spr-cx} \\ \text{MOTHER} \begin{bmatrix} \text{VAL.SPR} & \langle\rangle \end{bmatrix} \\ \text{HD-DTR} \quad \boxed{2} \begin{bmatrix} \text{VAL} \begin{bmatrix} \text{SPR} & \langle \boxed{1} \rangle \\ \text{COMPS} & \langle\rangle \end{bmatrix} \end{bmatrix} \\ \text{DTRS} \quad \langle \boxed{1}, \boxed{2} \rangle \end{bmatrix}$$

Principles Are Signs

- Head Feature Principle

$$\begin{bmatrix} \text{MOTHER.SYN.HEAD} & \boxed{1} \\ \text{HD-DTR.SYN.HEAD} & \boxed{1} \end{bmatrix}$$

- Semantic Inheritance Principle

$$\begin{bmatrix} \text{MOTHER.SEM} & \begin{bmatrix} \text{MODE} & \boxed{1} \\ \text{INDEX} & \boxed{2} \end{bmatrix} \\ \text{HD-DTR.SEM} & \begin{bmatrix} \text{MODE} & \boxed{1} \\ \text{INDEX} & \boxed{2} \end{bmatrix} \end{bmatrix}$$

- Valence Principle

$$\begin{bmatrix} \text{MOTHER.SYN.VAL} & / \boxed{1} \\ \text{HD-DTR.SYN.VAL} & / \boxed{1} \end{bmatrix}$$

The Root Condition (Start Symbol) Is a Sign

$$\begin{bmatrix} \text{SYN} & \begin{bmatrix} \text{HEAD} & \begin{bmatrix} \textit{verb} \\ \text{FORM} \;\; \textit{fin} \end{bmatrix} \\ \text{VAL} & \begin{bmatrix} \text{COMPS} & \langle\rangle \\ \text{SPR} & \langle\rangle \end{bmatrix} \\ \text{GAP} & \langle\rangle \end{bmatrix} \end{bmatrix}$$

Sign-Based Construction Grammar

- SBCG makes the analyses more uniform
- Puts information just where it is needed
- Merges the insights of HPSG and Construction Grammar
- Most recent work on HPSG is effectively work on SBCG

Look at Ian Sag's slides if you want to know more!

Acknowledgments and References

- Course design and slides borrow heavily from Emily Bender's course: *Linguistics 566: Introduction to Syntax for Computational Linguistics*
  http://courses.washington.edu/ling566

- This lecture complements Sag et al. (2003, Chapter 16)

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References

Ivan A. Sag, Tom Wasow, and Emily Bender. 2003. *Syntactic Theory: A Formal Introduction*. CSLI Publications, Stanford, second edition.